Problem 2. Write down the terms up to order 3 (i.e. $0 \le |\alpha| \le 3$) of the Taylor series about the origin ($\underline{x}^0 = \underline{0}$) of

(1) $$f(x,y) = \sin(2x+y).$$

Check your answer by using the fact that the first two nonzero terms of the Taylor series of $\sin(x)$ is $x - \frac{x^3}{3!}$.

Sol-n: Let $\underline{x} = \begin{bmatrix} x \\ y \end{bmatrix} \overset{\text{or}}{=} \langle x, y \rangle$ & $\underline{x}^0 = \underline{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \overset{\text{or}}{=} \langle 0, 0 \rangle$. (vectors).

Recall the notation for multidimensional Taylor series:

(2) $\alpha := (\alpha_1, \alpha_2, \dots \alpha_n)$ symbol (not a vector) for $n=2$ in $\mathbb{R}^2$, $n=3$ in $\mathbb{R}^3$ .... Then

how many derivatives

no multiplications here composition of operators $D_i^{\alpha_i}$ operating on fun-n f:

(3) $$D^\alpha f := D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n} f$$

with res. to 1st indep. var. — wrt 2nd indep. var. — wrt n^{th} indep. variable.

$$D_1^2 D_2^3 D_3^1 f = \frac{\partial^6 f(x,y,z)}{\partial x^2 \partial y^3 \partial z^1}$$

(4) $$\underline{x}^\alpha := x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdot \dots \cdot x_n^{\alpha_n}$$ where $\underline{x} = \langle x_1, x_2 \dots x_n \rangle$ vector.

(5) $$\alpha! := \alpha_1! \cdot \alpha_2! \cdot \dots \cdot \alpha_n! \qquad \alpha_i \in \mathbb{N} \quad 1, 2, 3, ..$$

(6) $$|\alpha| := \alpha_1 + \alpha_2 + \dots + \alpha_n.$$

Then Taylor series of $f(\underline{x})$ about $\underline{x} = \underline{x}^0$

(7) $$\boxed{\sum_{|\alpha| \ge 0} \frac{D^\alpha f(\underline{x}^0)}{\alpha!} (\underline{x} - \underline{x}^0)^\alpha}$$

So, let me start applying this (7) into the f-n in (1):

We've been given that

(8) $\underline{x}^0 = (0, 0)$ (about the origin) &

(9) $0 \le |\alpha| \le 3$ (up to 3rd order)

(10) $\alpha = \underset{\alpha_1\ \alpha_2}{(0,0)}, \underset{\alpha_1\ \alpha_2}{(1,0)}, (0,1), (1,1), (2,0), (0,2), (2,1), (1,2), (3,0), \underset{\alpha_1\ \alpha_2}{(0,3)}.$

Then by (7), Taylor series of 3rd order:

$$\frac{1}{(0,0)!} D^{(0,0)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(0,0)} + \frac{1}{(1,0)!} D^{(1,0)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(1,0)} +$$

$$+ \frac{1}{(0,1)!} D^{(0,1)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(0,1)} + \frac{1}{(1,1)!} D^{(1,1)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(1,1)} + \frac{1}{(2,0)!} D^{(2,0)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(2,0)}$$

$$+ \frac{1}{(0,2)!} D^{(0,2)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(0,2)} + \frac{1}{(2,1)!} D^{(2,1)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(2,1)} + \frac{1}{(1,2)!} D^{(1,2)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(1,2)}$$

$$+ \frac{1}{(0,3)!} D^{(0,3)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(0,3)} + \frac{1}{(3,0)!} D^{(3,0)} f(\underline{0}) \cdot (\underline{x} - \underline{0})^{(3,0)}.$$

Note that by (3), (4), (5), (6), for symbol $\alpha = (2, 1)$

(11) $\alpha! = (2,1)! := 2! \cdot 1!$ by (5);

(12) $D^{\alpha} f(\underline{0}) = D^{(2,1)} f(\underline{0}) = D_{\boxed{1}} D_{\boxed{2}} f(\underline{0}) = \left.\dfrac{\partial^3 f}{\partial x^2 \partial y^1}\right|_{(0,0)} \underset{=}{\text{or}} \dfrac{\partial^3 f(0,0)}{\partial x^2 \partial y}$ by (3)

2 times deriv. 1 times deriv.

$x =$ 1st indep. var. $y =$ 2nd indep. var. $\langle 0,0 \rangle$

2 times derivative wrt 1st indep. var. $\boxed{D_1^2}$

1 times (once) derivative wrt 2nd indep. var. $\boxed{D_2^1}$

and finally,

(13) $(\underline{x} - \underline{0})^{\alpha} = \left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right)^{(2,1)} = \left(\begin{bmatrix} x-0 \\ y-0 \end{bmatrix}\right)^{(2,1)} = (x-0)^2 \cdot (y-0)^1$ by (4).

vectors $\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Putting (11), (12), (13) all together into our Taylor series gives

$$(14) \quad \sum_{|\alpha| \geq 0} \frac{D^{\alpha} f(\underline{x}^0)}{\alpha!} (\underline{x} - \underline{x}^0)^{\alpha} =$$

$$= \frac{1}{0! \cdot 0!} D^{(0,0)} f(0,0) \cdot (x-0)^0 \cdot (y-0)^0 +$$

$$+ \frac{1}{1!\, 0!} D^{(1,0)} f(0,0) \cdot (x-0)^1 \cdot (y-0)^0 +$$

$$+ \frac{1}{0! \cdot 1!} D^{(0,1)} f(0,0) \cdot (x-0)^0 \cdot (y-0)^1 +$$

$$+ \frac{1}{1! \cdot 1!} D^{(1,1)} f(0,0) \cdot (x-0)^1 \cdot (y-0)^1 +$$

$$+ \frac{1}{2! \cdot 0!} D^{(2,0)} f(0,0) \cdot (x-0)^2 \cdot (y-0)^0 + \frac{1}{0! \cdot 2!} \cdot D^{(0,2)} f(0,0) \cdot (x-0)^0 \cdot (y-0)^2 +$$

$$+ \frac{1}{2! \cdot 1!} \cdot D^{(2,1)} f(0,0) \cdot (x-0)^2 \cdot (y-0)^1 + \frac{1}{1! \cdot 2!} \cdot D^{(1,2)} f(0,0) \cdot (x-0)^1 \cdot (y-0)^2 +$$

$$+ \frac{1}{0! \cdot 3!} \cdot D^{(0,3)} f(0,0) \cdot (x-0)^0 \cdot (y-0)^3 + \frac{1}{3! \cdot 0!} D^{(3,0)} f(0,0) \cdot (x-0)^3 \cdot (y-0)^0 .$$

Now let me compute each term separately for

$$f(x,y) = \sin(2x+y)$$

$$(15) \quad D^{(0,0)} f(0,0) = \frac{\partial^{0+0} f(0,0)}{\partial x^0 \, \partial y^0} \underset{\text{deriv.}}{=} f(0,0) = \sin(2 \cdot 0 + 0) = 0.$$

$$(15') \quad D^{(1,0)} f(0,0) = \frac{\partial^{1+0} f(0,0)}{\partial x^1 \, \partial y^0} = \frac{\partial f(0,0)}{\partial x} = \frac{\partial}{\partial x}\left[\sin(2x+y)\right]_{(0,0)} = 2\cos(2x+y)\Big|_{(0,0)} = 2.$$

(15'') $D^{(0,1)}f(0,0) = \left.\dfrac{\partial^{0+1} f}{\partial x^0\, \partial y^1}\right|_{(0,0)} = \dfrac{\partial}{\partial y}\Big[\sin(2x+y)\Big]_{(0,0)} = \cos(2x+y)\Big|_{(0,0)} =$

$= \cos(2\cdot 0+0) = 1.$

(16) $D^{(1,1)}f(0,0) = \dfrac{\partial^{1+1} f(0,0)}{\partial x^1\, \partial y^1} = \dfrac{\partial^2 f(0,0)}{\partial x \partial y} = \dfrac{\partial}{\partial x}\left[\dfrac{\partial}{\partial y}\Big(\sin(2x+y)\Big)\right]_{(0,0)} =$

$= \dfrac{\partial}{\partial x}\Big[\cos(2x+y)\Big]_{(0,0)} = \Big[-2\cdot\sin(2x+y)\Big]_{(0,0)} = -2\sin(0+0) = 0.$

(17) $D^{(2,0)}f(0,0) = \dfrac{\partial^{2+0} f(0,0)}{\partial x^2\, \partial y^0} = \dfrac{\partial^2 f(0,0)}{\partial x^2} = \dfrac{\partial}{\partial x}\left[\dfrac{\partial}{\partial x}\Big(\sin(2x+y)\Big)\right]_{(0,0)}$

$= \dfrac{\partial}{\partial x}\Big[2\cos(2x+y)\Big]_{(0,0)} = \Big[-2\cdot 2\sin(2x+y)\Big]_{(0,0)} = -4\cdot\sin(2\cdot 0+0) = 0.$

(18) $D^{(0,2)}f(0,0) = \dfrac{\partial^{0+2} f(0,0)}{\partial x^0\, \partial y^2} = \dfrac{\partial^2 f(0,0)}{\partial y^2} = \dfrac{\partial}{\partial y}\left[\dfrac{\partial}{\partial y}\Big(\sin(2x+y)\Big)\right]_{(0,0)} =$

$= \dfrac{\partial}{\partial y}\Big[\cos(2x+y)\Big]_{(0,0)} = \Big[-\sin(2x+y)\Big]_{(0,0)} = -\sin(2\cdot 0+0) = 0.$

by the result of (16)

(19) $D^{(2,1)}f(0,0) = \dfrac{\partial^{2+1} f(0,0)}{\partial x^2\, \partial y^1} = \dfrac{\partial^3 f(0,0)}{\partial x^2 \partial y} = \dfrac{\partial}{\partial x}\left[\dfrac{\partial^2 f}{\partial x \partial y}\right]_{(0,0)} = \dfrac{\partial}{\partial x}\Big[-2\sin(2x+y)\Big]_{(0,0)} =$

$= \Big[2\cdot(-2)\cos(2x+y)\Big]_{(0,0)} = -4\cos(2\cdot 0+0) = -4.$

(20) $$D^{(1,2)}f(0,0) = \frac{\partial^{1+2} f(0,0)}{\partial x^1 \partial y^2} = \frac{\partial^3 f(0,0)}{\partial x \partial y^2} = \frac{\partial}{\partial x}\left[\frac{\partial^2 f(0,0)}{\partial y^2}\right] \overset{\text{by (18)}}{=}$$

$$= \frac{\partial}{\partial x}\left[-\sin(2x+y)\right]_{(0,0)} = \left[-2\cos(2x+y)\right]_{(0,0)} = -2\cos(0) = -2.$$

(21) $$D^{(0,3)}f(0,0) = \frac{\partial^{0+3} f(0,0)}{\partial x^0 \partial y^3} = \frac{\partial^3 f(0,0)}{\partial y^3} = \frac{\partial}{\partial y}\left[\frac{\partial^2 f(0,0)}{\partial y^2}\right] \overset{\text{by (18)}}{=}$$

$$= \frac{\partial}{\partial y}\left[-\sin(2x+y)\right]_{(0,0)} = \left[-1\cos(2x+y)\right]_{(0,0)} = -1\cos(2\cdot 0+0) = -1.$$

(22) $$D^{(3,0)}f(0,0) = \frac{\partial^{3+0} f(0,0)}{\partial x^3 \partial y^0} = \frac{\partial^3 f(0,0)}{\partial x^3} = \frac{\partial}{\partial x}\left[\frac{\partial^2 f}{\partial x^2}\right] \overset{\text{by (17)}}{=} \frac{\partial}{\partial x}\left[-4\sin(2x+y)\right]_{(0,0)} =$$

$$= \left[-8\cos(2x+y)\right]_{(0,0)} = -8\cos(2\cdot 0+0) = -8.$$

So, substituting all entities (15) – (22) back into (14) gives

$$\sum_{|\alpha|\geq 0} \frac{D^\alpha f(\underline{0})}{\alpha!}(\underline{x}-\underline{0})^\alpha = \frac{(0)}{0!\cdot 0!}\cdot(x-0)^0\cdot(y-0)^0 + \frac{(2)}{1!\cdot 0!}(x-0)^1\cdot(y-0)^0 +$$

$$+ \frac{(1)}{0!\,1!}\cdot(x-0)^0\cdot(y-0)^1 + \frac{(0)}{1!\cdot 1!}\cdot(x-0)^1\cdot(y-0)^1 + \frac{(0)}{2!\cdot 0!}\cdot(x-0)^2\cdot(y-0)^0 +$$

$$+ \frac{(0)}{0!\cdot 2!}\cdot(x-0)^0\cdot(y-0)^2 + \frac{(-4)}{2!\cdot 1!}\cdot(x-0)^2\cdot(y-0)^1 +$$

$$+ \frac{(-2)}{1!\cdot 2!}\cdot(x-0)^1\cdot(y-0)^2 + \frac{(-1)}{0!\cdot 3!}\cdot(x-0)^0\cdot(y-0)^3 + \frac{(-8)}{3!\cdot 0!}\cdot(x-0)^3\cdot(y-0)^0$$

$$= 2x + y - 2x^2\cdot y - xy^2 - \frac{y^3}{6} - \frac{4x^3}{3}.$$

So

$$(23) \quad \sum_{0 \le |\alpha| \le 3} \frac{D^{\alpha} f(\underline{0})}{\alpha!} (\underline{x} - \underline{0})^{\alpha} = 2x + y - 2x^2 y - xy^2 - \frac{y^3}{6} - \frac{4x^3}{3}$$

Checking (22):

By Taylor series expansion formula of $\sin x = x - \frac{x^3}{3!}$,

$$\sin(2x+y) = (2x+y) - \frac{(2x+y)^3}{3!} = (2x+y) -$$

$$- \frac{1}{6}\left[8x^3 + 3\cdot(4x^2)\cdot y + 3\cdot(2x)\cdot y^2 + y^3\right] =$$

$$(24) \quad = \boxed{(2x+y) - 2x^2 y - xy^2 - \frac{y^3}{6} - \frac{4x^3}{3}} \quad \checkmark$$

which, of course, confirms the result in (23). $\blacksquare$